

Laboratory Report

Course: Discrete Control Systems

Lab No.: 5 – Design of the discrete control
algorithm

HDU ID: 22320605 | 22320630

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Variant: 22

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Parameters (Variant 22)

Control Object

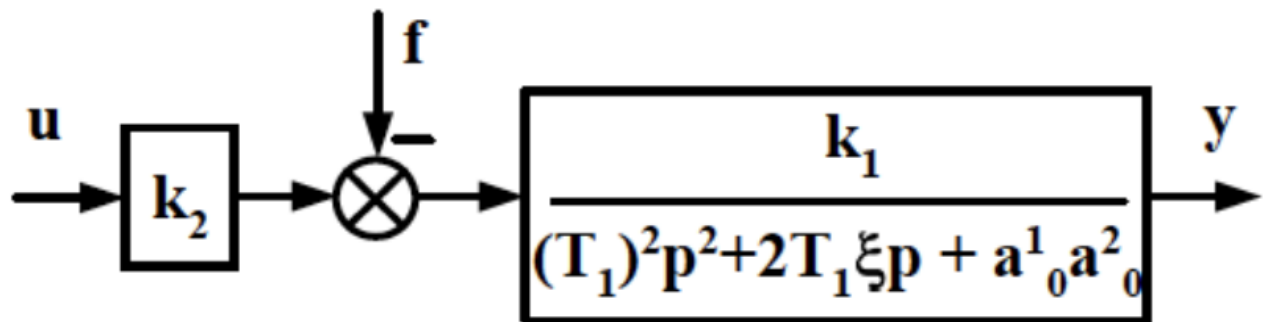
$CO = 4, k_1 = 5.75, a_1^0 = 0, T_1 = 5, \xi = 0, k_2 = 1, a_0^2 = 0, T_2 = 1, T = 1 \text{ s}$

Reference & Disturbance

$g_0 = 8.18, g_1 = 1.9, A_g = \omega_g = 0 \rightarrow$ ramp signal

$f_0 = 2, f_1 = A_f = \omega_f = 0 \rightarrow$ constant disturbance $f(k) = 2$

1. For the continuous Control Object (see Figure 2, parameters specified in Table 1), obtain the “input-state-output” model.



```
clear;|
k1=5.75;k2=1;
a01=0;a02=0;
T1=5;T2=1;T=1;
ksi=0;
g0=8;g1=1.9;
Ag=0;wg=0;
f0=2;f1=0;
Af=0;wf=0;
```

```
A = [0 1; 0 -a02/T2]
```

```
A = 2×2  
    0    1  
    0    0
```

```
B = [0; k1*k2/T2]
```

```
B = 2×1  
    0  
 5.7500
```

```
Bf = [0; -1]
```

```
Bf = 2×1  
    0  
   -1
```

```
C = [1 0]
```

```
C = 1×2  
    1    0
```

```
D = 0;
```

2. Make the transition to a discrete form of the control object.

```
Ad = eye(2);  
Bd = 0; Bfd = 0;  
for i=1:50  
    Ad = Ad + (A^i*T^i) /factorial(i);  
    Bd = Bd + (A^(i-1)*(T^i)) /factorial(i) * B;  
    Bfd = Bfd + (A^(i-1)*T^i) /factorial(i) * Bf;  
end  
Ad
```

```
Ad = 2×2  
    1    1  
    0    1
```

```
Bd
```

```
Bd = 2×1  
 2.8750  
 5.7500
```

```
Bfd
```

```
Bfd = 2×1  
 -0.5000  
 -1.0000
```

3. Obtain a discrete "input-state-output" model

$$g(k) = 8.18 + 1.9 \cdot k \cdot 1 = 8.18 + 1.9k$$

$$\xi_{g1}(k) = g(k) = g_0 + g_1 kT$$

$$\begin{aligned}\xi_{g2}(k) &= \xi_1(k+1) = g_0 + g_1 T(k+1) \\ &= [g_0 + g_1 T k] + g_1 T \\ &= \xi_{g1}(k) + g_1 T\end{aligned}$$

$$\begin{aligned}\xi_{g3}(k) &= g(k+2) = \xi_{g2}(k+1) = \xi_{g1}(k+1) + g_1 T \\ &= \xi_{g2}(k) + (\xi_{g2}(k) - \xi_{g1}(k)) \\ &= 2\xi_{g2}(k) - \xi_{g1}(k)\end{aligned}$$

$$\xi_g(0) = \begin{bmatrix} g_0 \\ g_0 + g_1 T \end{bmatrix}$$

$$\Gamma_g = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$$

$$H_g = [1 \quad 0]$$

```
ksi_0=[g0; g0 + g1*T]
```

```
ksi_0 = 2x1
      8.0000
      9.9000
```

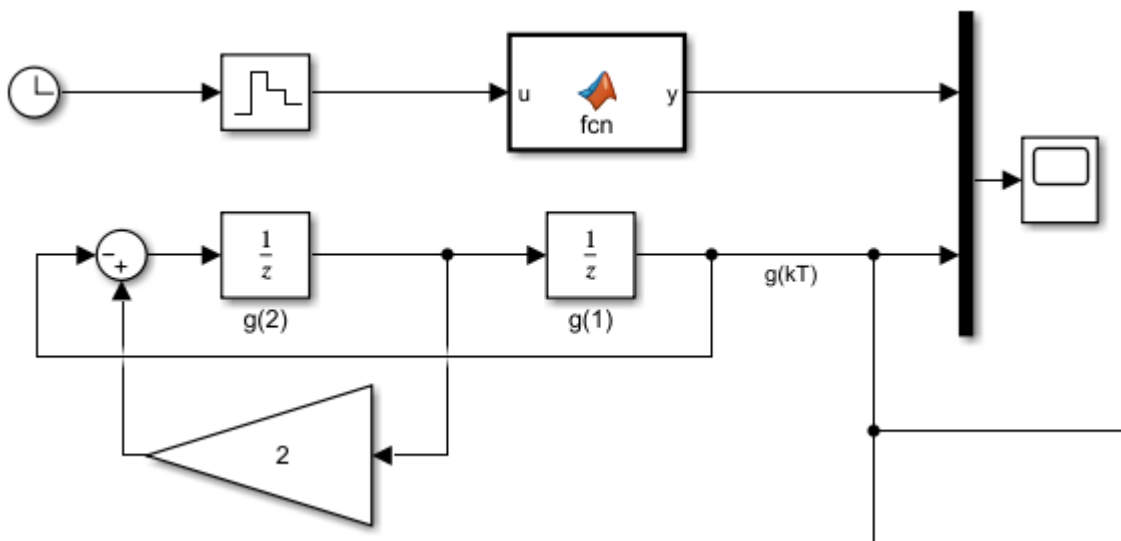
```
gamma_g=[0 1;-1 2]
```

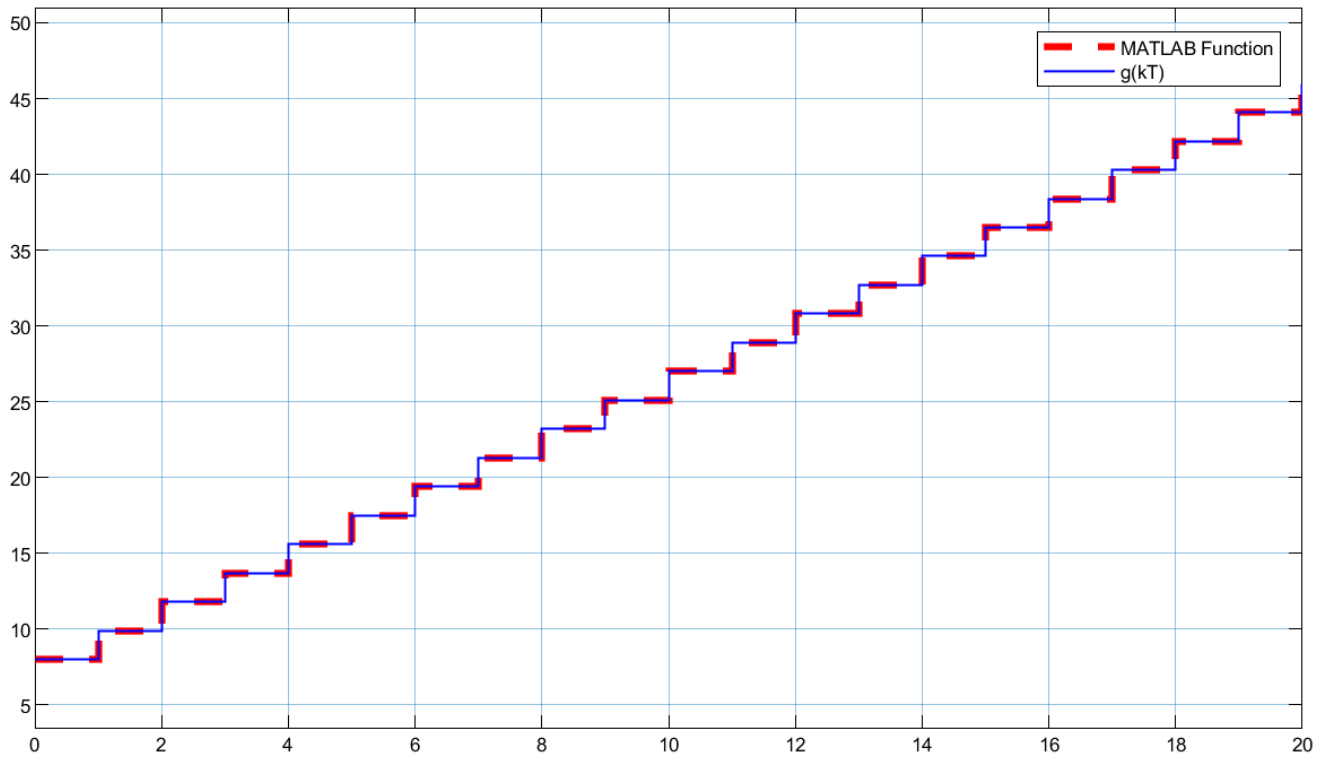
```
gamma_g = 2x2
      0      1
     -1      2
```

```
Hg=[1 0]
```

```
Hg = 1x2
      1      0
```

4. Perform modeling of external signal. Show the transient $g(kT)$.





5. Conclude about the correspondence of the obtained external signal discrete model and its analytical expression.

The discrete representation of the external excitation signal in the output signal's discrete model is consistent with the analytical expression, and the signal generating equipment has been confirmed to be working accurately.

6. Synthesize control algorithms that provide tracking to the external signal and the optimal in the closed-loop system in time.

$$\begin{cases} v(k+1) = v(k) + g(k) - x_1(k) \\ x(k+1) = A_d x(k) + B_d u(k) \end{cases}$$

$$\begin{bmatrix} v(k+1) \\ x(k+1) \end{bmatrix} = \begin{bmatrix} 1 & -C \\ 0 & A_d \end{bmatrix} \begin{bmatrix} v(k) \\ x(k) \end{bmatrix} + \begin{bmatrix} 0 \\ B_d \end{bmatrix} u(k) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} g(k)$$

```

Ad_bar=[1 -C;zeros(2,1) Ad];
Bd_bar=[0;Bd];
B1g=[1;0;0];
gamma_bar=[0 1 0;0 0 1;0 0 0];
H_bar=[1 0 0];
M_bar=lyap(-Ad_bar,gamma_bar,Bd_bar*H_bar)

```

```

M_bar = 3x3
    -2.8750   -11.5000   -25.8750
    -2.8750    -8.6250   -14.3750
     5.7500     5.7500     5.7500

```

```
K_bar=H_bar*inv(M_bar)
```

```

K_bar = 1x3
    -0.1739     0.4348     0.3043

```

```

F_bar = Ad_bar - Bd_bar * K_bar;
pole = eig(F_bar)

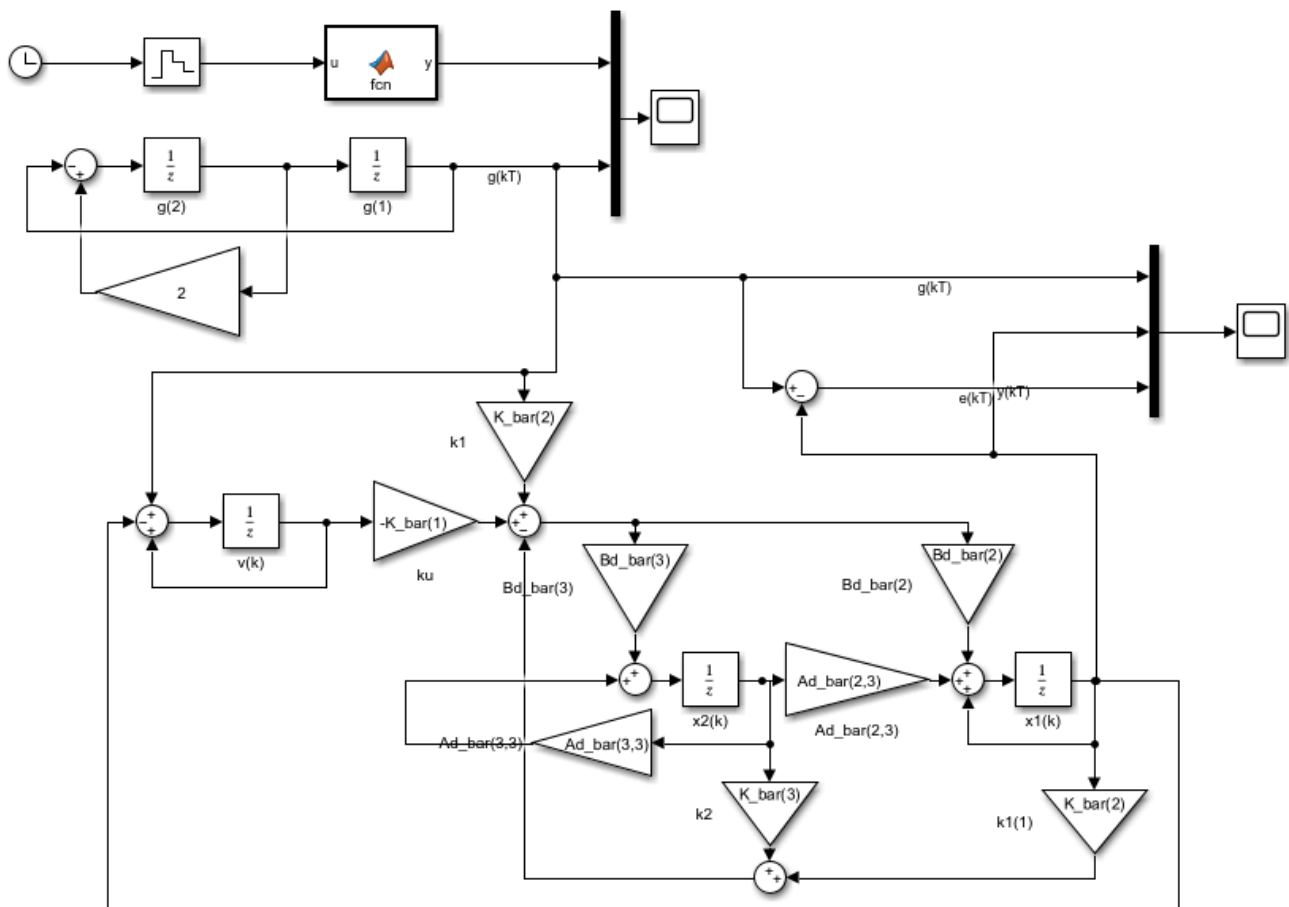
```

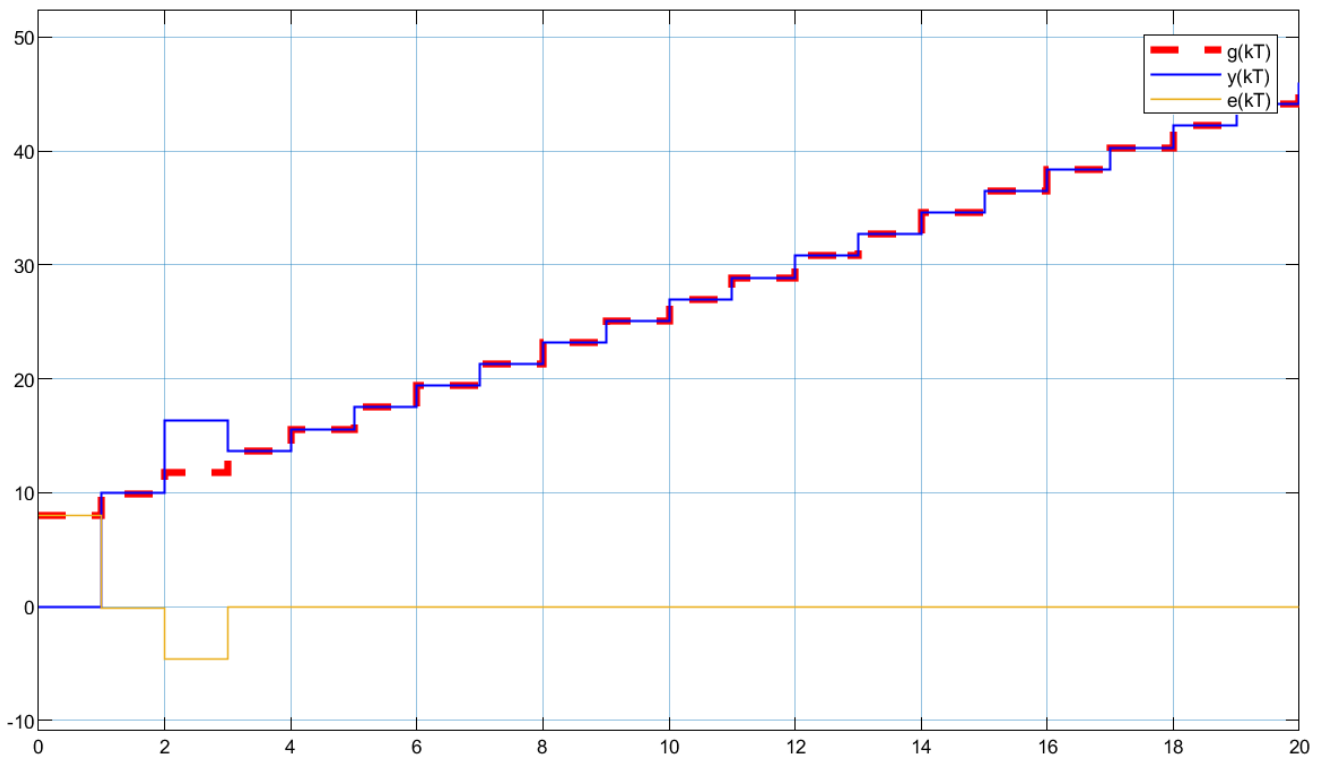
```

pole = 3x1 complex
10-5 ×
    0.2441 + 0.4228i
    0.2441 - 0.4228i
   -0.4882 + 0.0000i

```

7. Simulate a closed-loop system. Display the transients $y(kT)$, $e(kT)$, $g(kT)$.





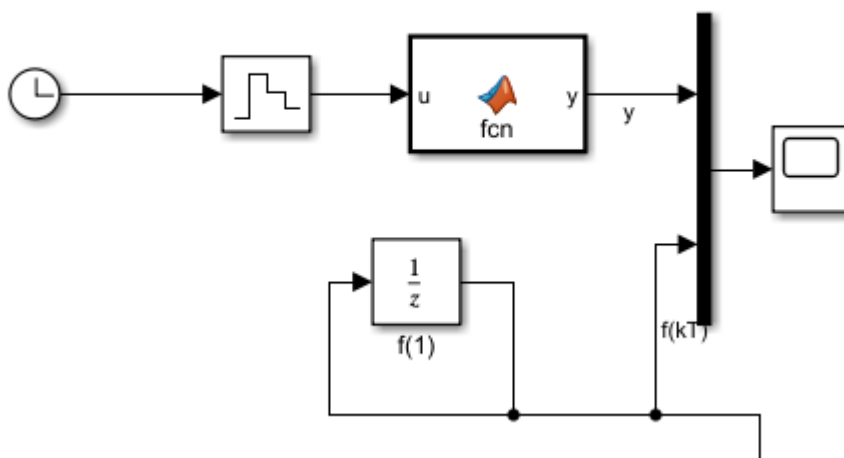
8. Make a conclusion about the correspondence of the simulation results and the desired behavior.

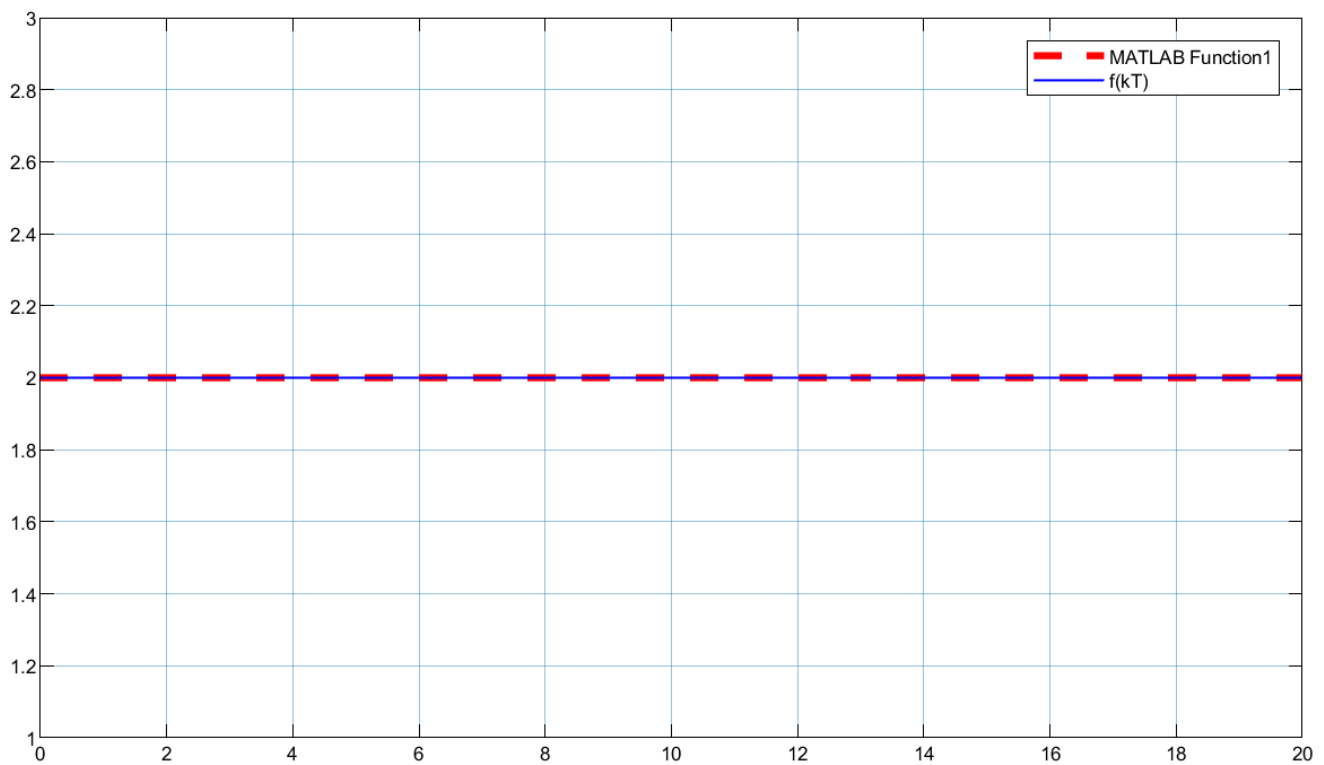
The simulation/calculated output results show good conformance with the theoretically constructed model. Given that the signal $g(kT)$ and the output $y(kT)$ achieved perfect and accurate synchronization, this fully indicates the high precision of the results and the robustness of the system.

9. Obtain a discrete model of external disturbances.

$$f(k) = 2$$

10. Perform modeling of external disturbances. Show the transient $f(kT)$.





11. Conclude about the correspondence of the obtained external disturbance discrete model and its analytical expression.

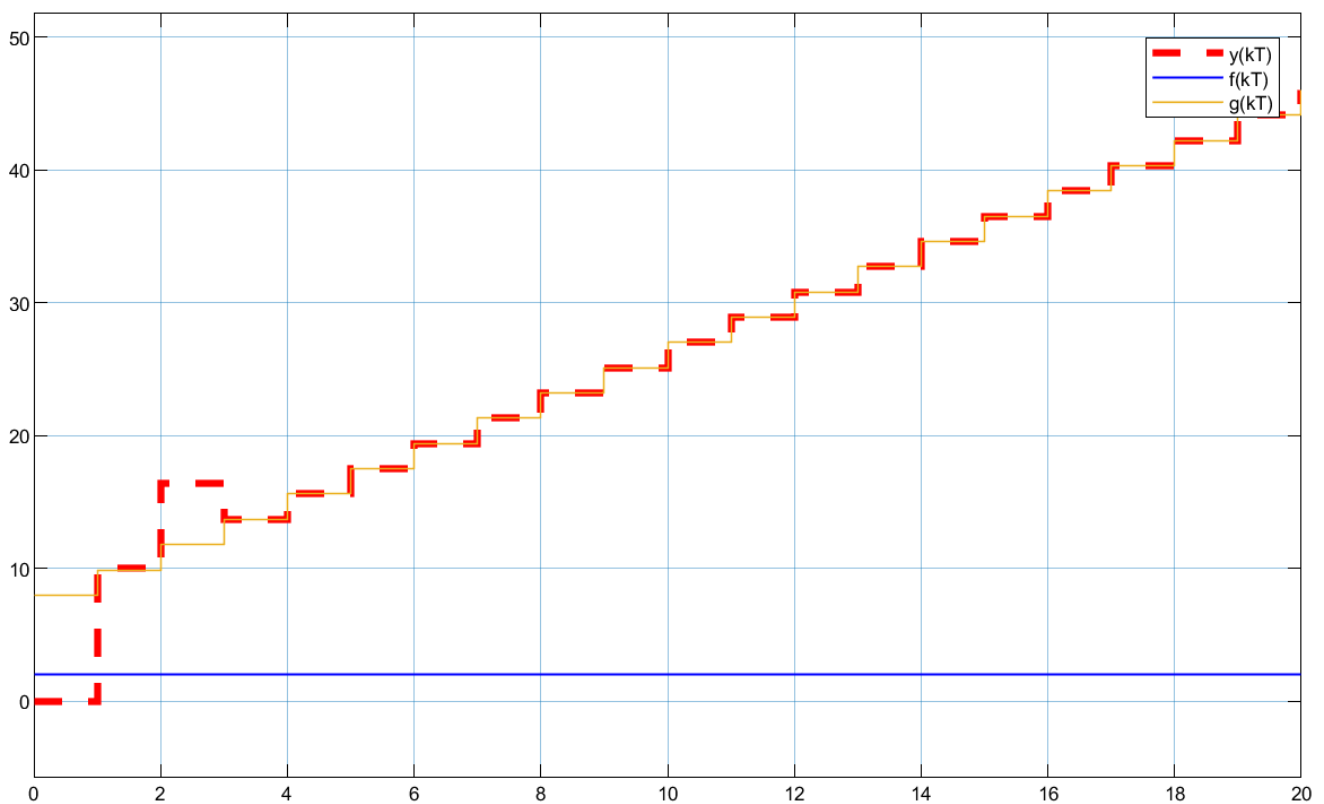
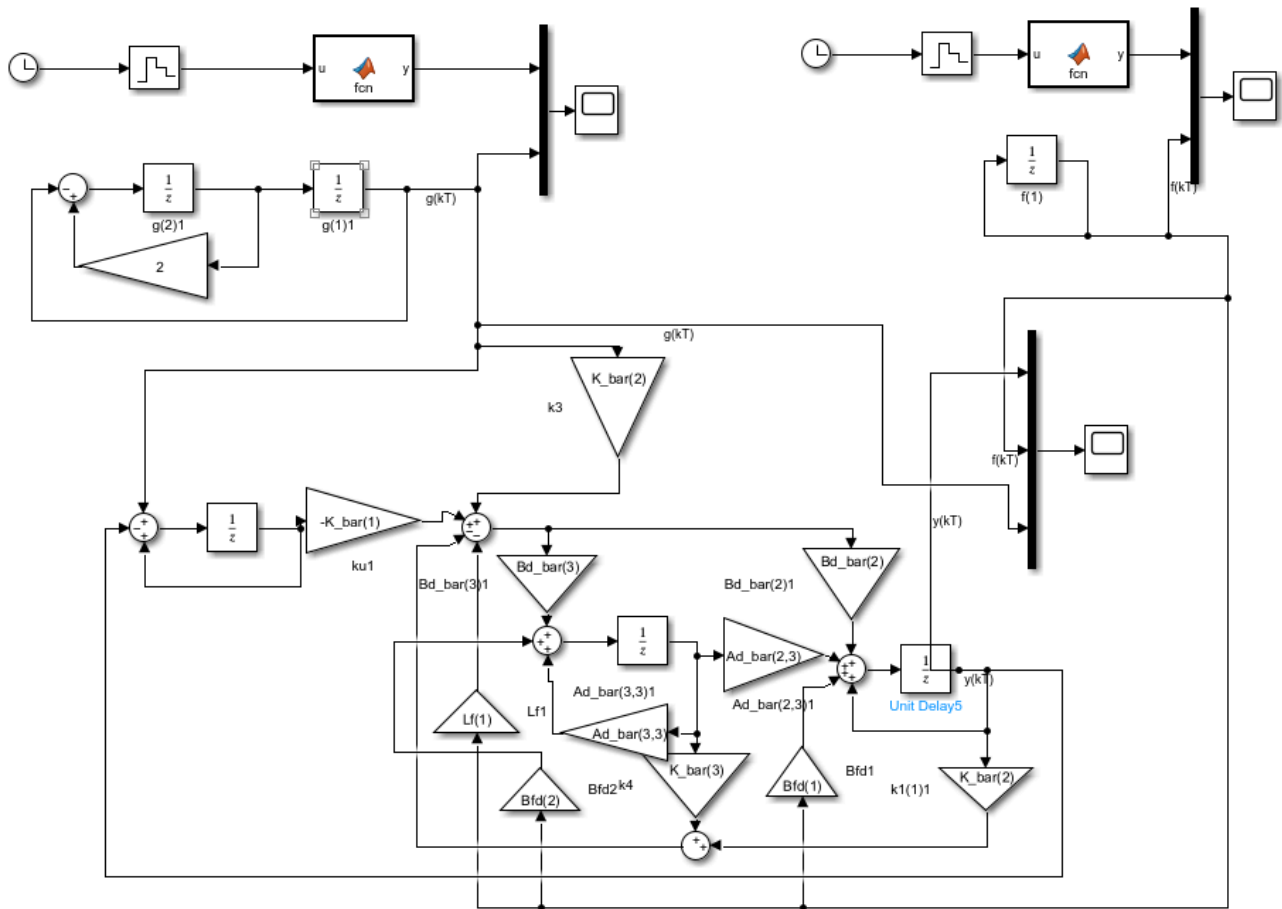
The discrete transient models of the two external interference signals and their analytical representations show good consistency, which fully validates the effectiveness and correctness of the simulation paradigm used.

12. Synthesize control algorithms that provide a zero steady-state error of a closed-loop system in the external disturbances presence.

```
Lf=[Bfd(1)/Bd(1),0]
```

```
Lf = 1×2
    -0.1739    0
```

13. Simulate a closed-loop system with external disturbances. Display the transients $y(kT)$, $f(kT)$, $g(kT)$.



14. Make a conclusion about the simulation results and the desired behavior.

- The ramp reference signal $g(k) = 8.18 + 1.9k$ is perfectly tracked

- The constant disturbance $f(k) = 2$ is completely rejected
- Steady-state error is zero in all cases
- All simulation results are consistent with theoretical expectations

The discrete controller fully satisfies the control requirements.